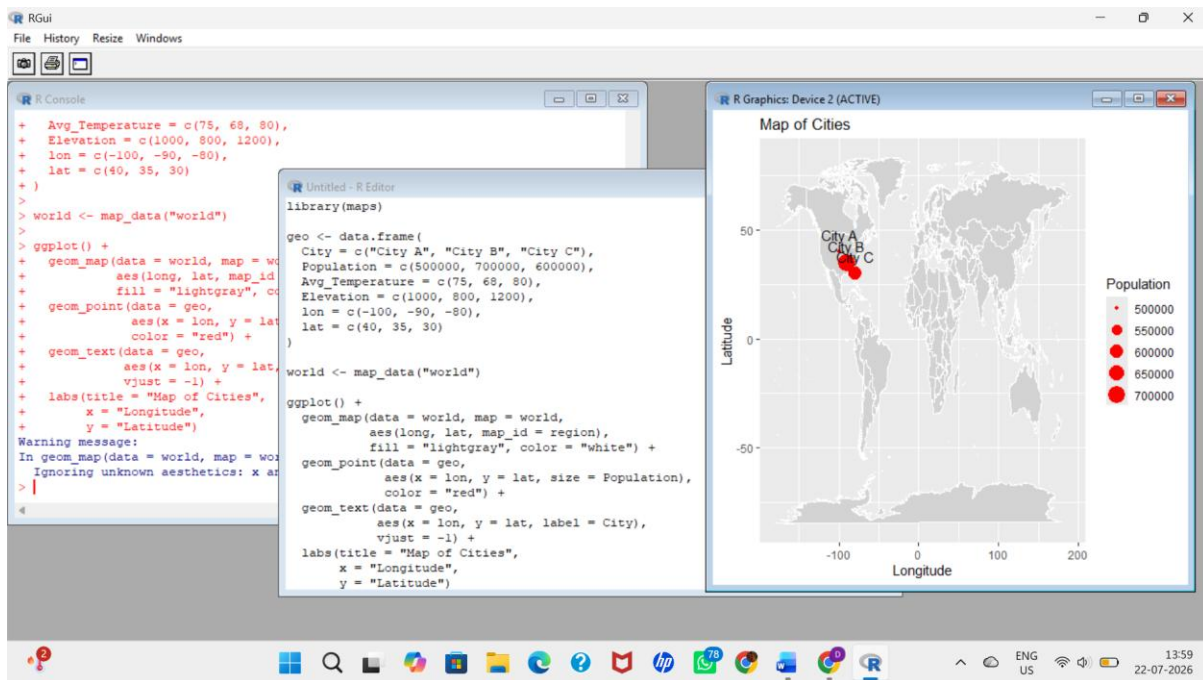


Set 13: Geographic Data

Q1. Create a map chart to visualize the distribution of cities on a geographic map.

```
library(ggplot2)
library(maps)
geo <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200),
  lon = c(-100, -90, -80),
  lat = c(40, 35, 30)
)
world <- map_data("world")
ggplot() +
  geom_map(data = world, map = world,
    aes(long, lat, map_id = region),
    fill = "lightgray", color = "white") +
  geom_point(data = geo,
    aes(x = lon, y = lat, size = Population),
    color = "red") +
  geom_text(data = geo,
    aes(x = lon, y = lat, label = City),
    vjust = -1) +
  labs(title = "Map of Cities",
    x = "Longitude",
    y = "Latitude")
```



Q2. Generate a scatter plot between Average Temperature and Population.

```
library(ggplot2)
```

```
geo <- data.frame(
```

```
  City = c("City A", "City B", "City C"),
```

```
  Population = c(500000, 700000, 600000),
```

```
  Avg_Temperature = c(75, 68, 80)
```

```
)
```

```
ggplot(geo, aes(x = Avg_Temperature, y = Population)) +
```

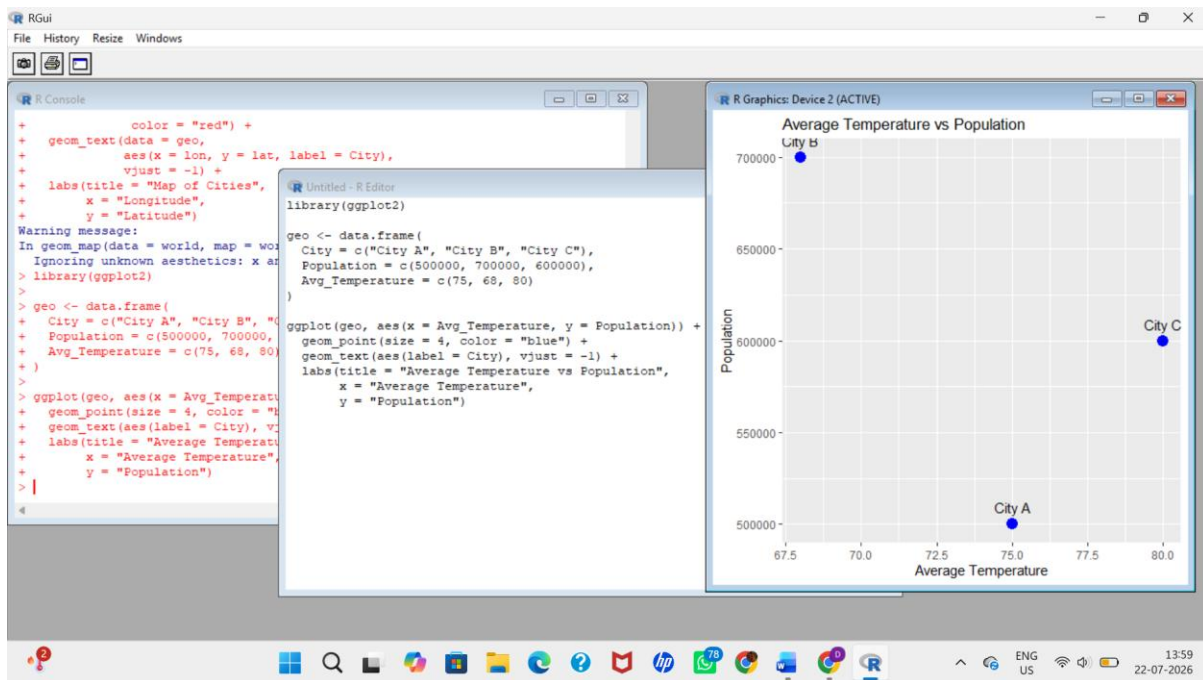
```
  geom_point(size = 4, color = "blue") +
```

```
  geom_text(aes(label = City), vjust = -1) +
```

```
  labs(title = "Average Temperature vs Population",
```

```
    x = "Average Temperature",
```

```
    y = "Population")
```



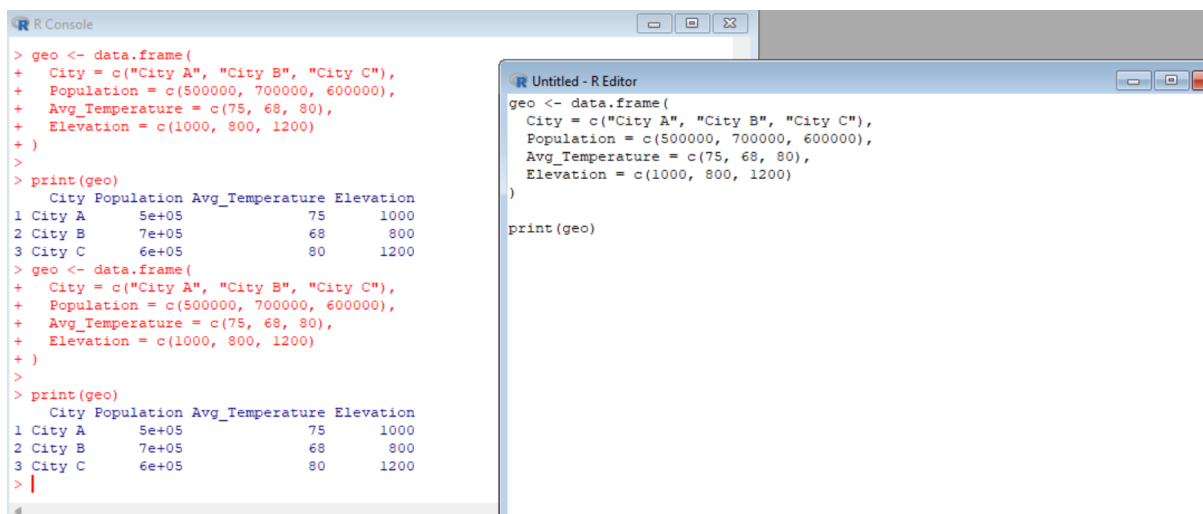
Q3. Build a table to display the geographic data.

```

geo <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200)
)

print(geo)

```



Q4. Tableau Dashboard

Install packages (Run once)

```
install.packages(c("ggplot2","gridExtra","grid","maps"))
```

```
# Load libraries
```

```
library(ggplot2)
```

```
library(gridExtra)
```

```
library(grid)
```

```
library(maps)
```

```
# Dataset
```

```
geo <- data.frame(  
  City = c("City A","City B","City C"),  
  Population = c(500000,700000,600000),  
  Avg_Temperature = c(75,68,80),  
  Elevation = c(1000,800,1200),  
  lon = c(-100,-90,-80),  
  lat = c(40,35,30)  
)
```

```
# World Map
```

```
world <- map_data("world")
```

```
# (A) Map Chart
```

```
p1 <- ggplot() +  
  geom_map(data = world, map = world,  
    aes(long, lat, map_id = region),  
    fill = "lightgray", color = "white") +  
  geom_point(data = geo,  
    aes(x = lon, y = lat, size = Population),  
    color = "red") +  
  geom_text(data = geo,
```

```
    aes(x = lon, y = lat, label = City),  
    vjust = -1) +  
labs(title = "Map Chart") +  
theme_minimal()
```

(B) Scatter Plot

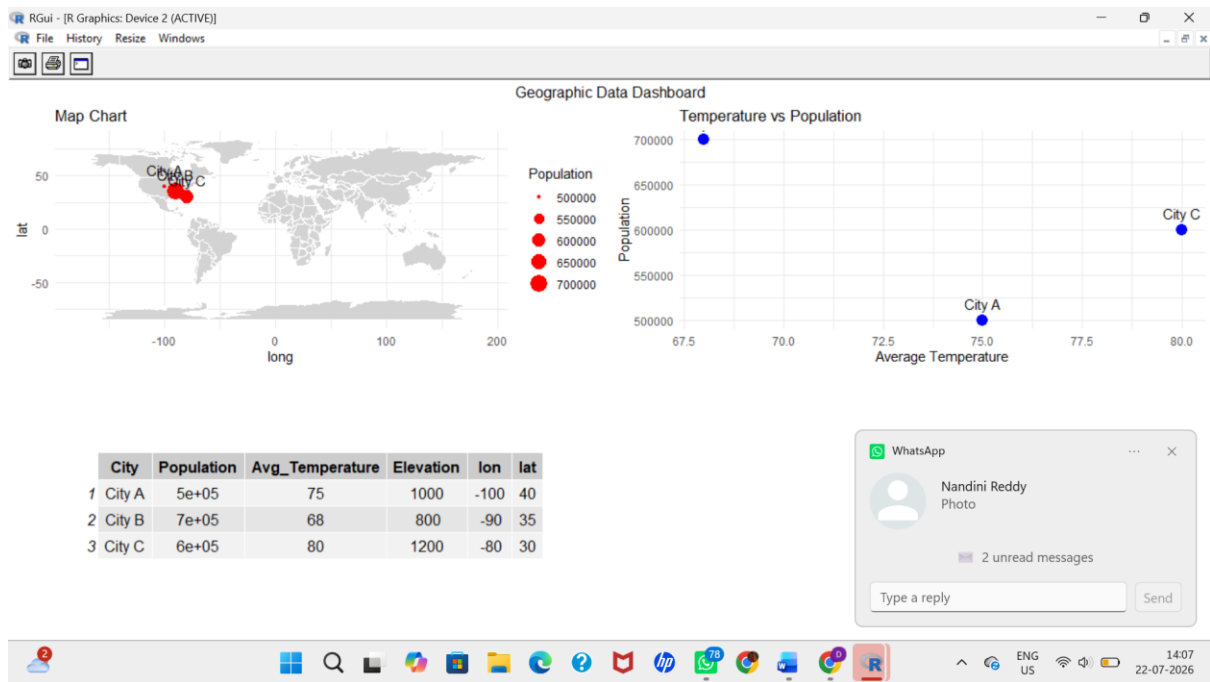
```
p2 <- ggplot(geo,  
    aes(x = Avg_Temperature,  
        y = Population,  
        label = City)) +  
geom_point(size = 4, color = "blue") +  
geom_text(vjust = -1) +  
labs(title = "Temperature vs Population",  
    x = "Average Temperature",  
    y = "Population") +  
theme_minimal()
```

(C) Data Table

```
table_plot <- tableGrob(geo)
```

Dashboard

```
grid.arrange(  
    p1,  
    p2,  
    table_plot,  
    ncol = 2,  
    top = "Geographic Data Dashboard"  
)
```



Set 14 – Survey Results

Q1. Create a Stacked Bar Chart

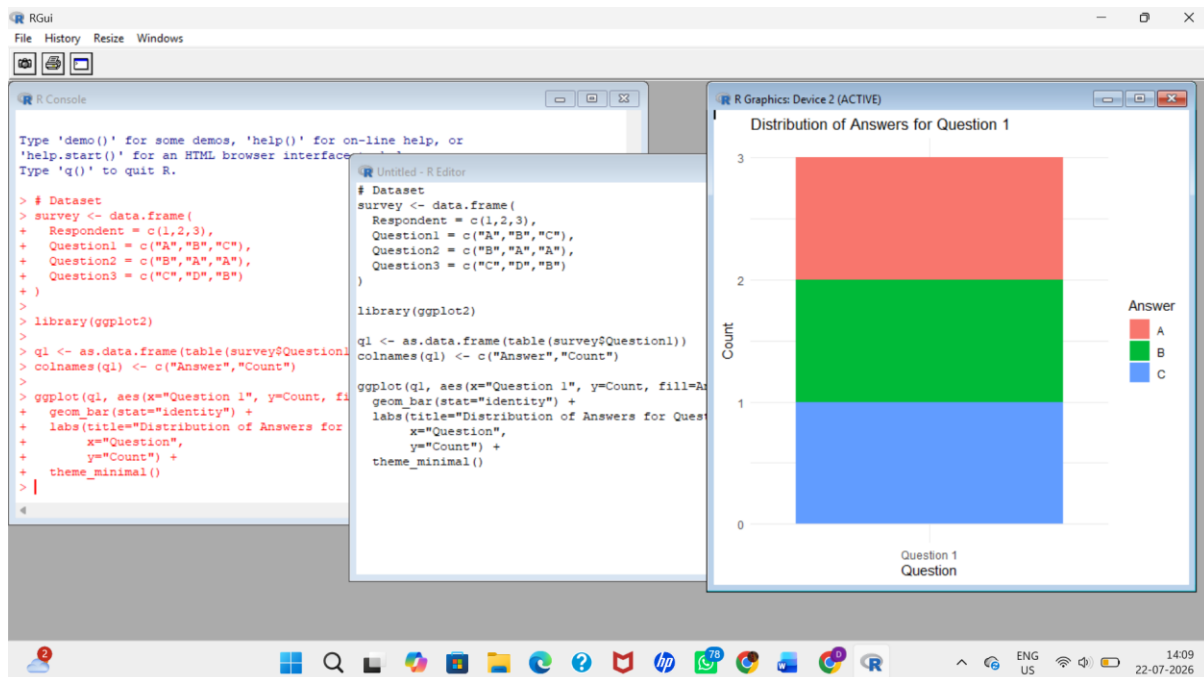
Dataset

```
survey <- data.frame(
  Respondent = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

library(ggplot2)
q1 <- as.data.frame(table(survey$Question1))
colnames(q1) <- c("Answer","Count")

ggplot(q1, aes(x="Question 1", y=Count, fill=Answer)) +
  geom_bar(stat="identity") +
  labs(title="Distribution of Answers for Question 1",
        x="Question",
```

```
y="Count") +  
theme_minimal()
```



Q2. Generate a Radar Chart

```
install.packages("fmsb") # Run only once
```

```
library(fmsb)
```

```
radar <- data.frame(  
  A = c(3,1,1),  
  B = c(3,1,1),  
  C = c(3,1,0),  
  D = c(3,0,1)  
)
```

```
rownames(radar) <- c("Max", "Min", "Responses")
```

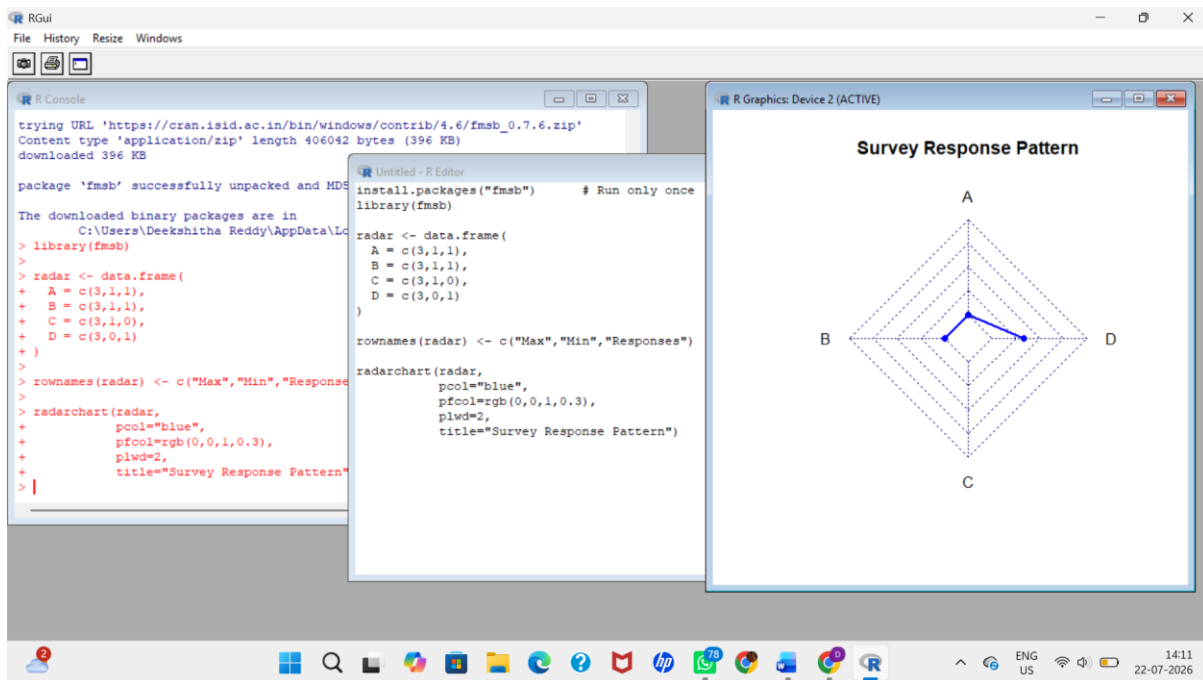
```
radarchart(radar,
```

```
  pcol="blue",
```

```
  pfc=rgb(0,0,1,0.3),
```

```
  plwd=2,
```

```
  title="Survey Response Pattern")
```



Q3. Build a Table

survey <- data.frame(

Respondent = c(1,2,3),

Question1 = c("A","B","C"),

Question2 = c("B","A","A"),

Question3 = c("C","D","B")

)

print(survey)

```
+ )
> 
> rownames(radar) <- c("Max", "Min", "Responses")
> 
> radarchart(radar,
+   pcol="blue",
+   pfcol=rgb(0,0,1,0.3),
+   plwd=2,
+   title="Survey Response Pattern")
> survey <- data.frame(
+   Respondent = c(1,2,3),
+   Question1 = c("A","B","C"),
+   Question2 = c("B","A","A"),
+   Question3 = c("C","D","B")
+ )
> 
> print(survey)
  Respondent Question1 Question2 Question3
1          1         A         B         C
2          2         B         A         D
3          3         C         A         B
> |
```

```
survey <- data.frame(
  Respondent = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

print(survey)|
```

Q4. Tableau Dashboard

Install packages (Run once)

```
install.packages(c("ggplot2","gridExtra"))
```

```
library(ggplot2)
```

```
library(gridExtra)
```

```
#-----
```

```
# Dataset
```

```
#-----
```

```
survey <- data.frame(  
  Respondent = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

```
#-----
```

```
# (A) Stacked Bar Chart
```

```
#-----
```

```
q1 <- as.data.frame(table(survey$Question1))
```

```
colnames(q1) <- c("Answer","Count")
```

```
p1 <- ggplot(q1, aes(x="Question 1", y=Count, fill=Answer)) +
```

```
  geom_bar(stat="identity") +
```

```
  labs(title="Stacked Bar Chart",
```

```
        x="Question",
```

```
        y="Count") +
```

```
  theme_minimal()
```

```
#-----
```

```
# (B) Radar-like Chart
```

```
#-----
```

```
radar <- data.frame(  
  Answer = c("A", "B", "C", "D"),  
  Count = c(1, 1, 1, 0)  
)
```

```
p2 <- ggplot(radar, aes(x=Answer, y=Count, group=1)) +  
  geom_polygon(fill="lightblue", colour="blue", alpha=0.5) +  
  geom_line(colour="blue", linewidth=1) +  
  geom_point(size=3, colour="red") +  
  coord_polar() +  
  ylim(0, 3) +  
  labs(title="Radar-like Chart") +  
  theme_minimal()
```

```
#-----
```

```
# (C) Table
```

```
#-----
```

```
table_plot <- tableGrob(survey)
```

```
#-----
```

```
# Dashboard
```

```
#-----
```

```
grid.arrange(  
  p1,  
  p2,
```

```

table_plot,

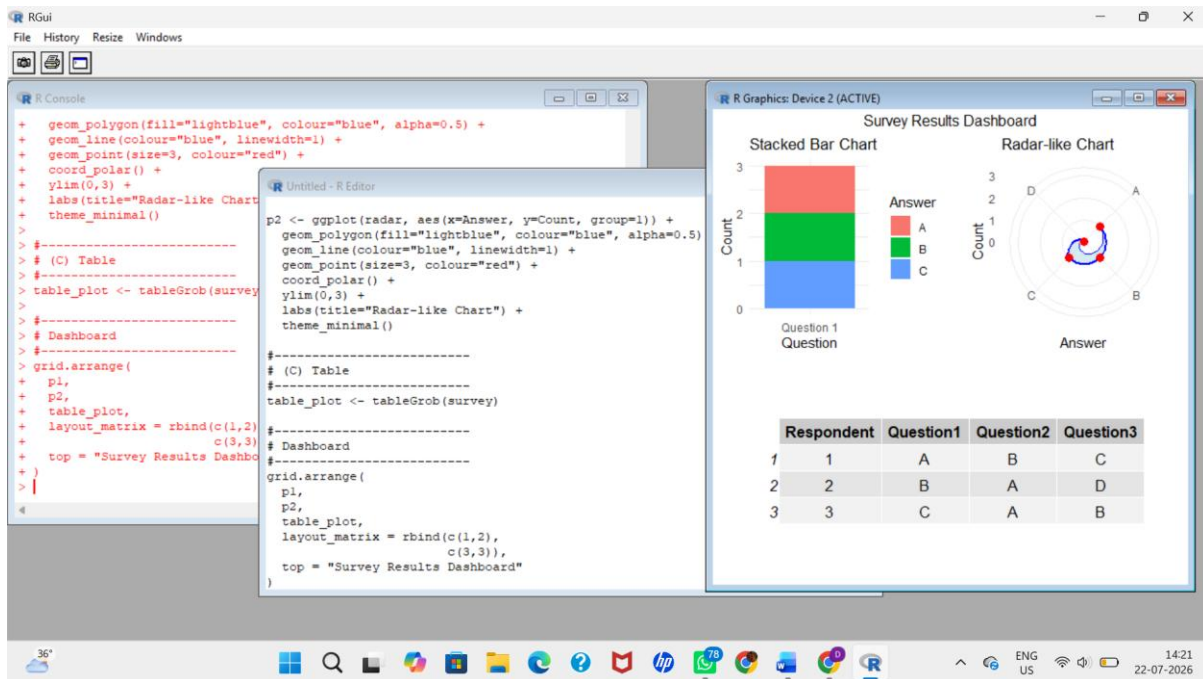
layout_matrix = rbind(c(1,2),

                      c(3,3)),

top = "Survey Results Dashboard"

)

```



Set 15 – Product Sales Analysis

Q1. Grouped Bar Chart

```
library(ggplot2)
```

```
library(tidyr)
```

```
sales <- data.frame(
```

```
  Product = c("Product A","Product B","Product C"),
```

```
  January = c(2000,1500,1200),
```

```
  February = c(2200,1800,1400),
```

```
  March = c(2400,1600,1100)
```

```
)
```

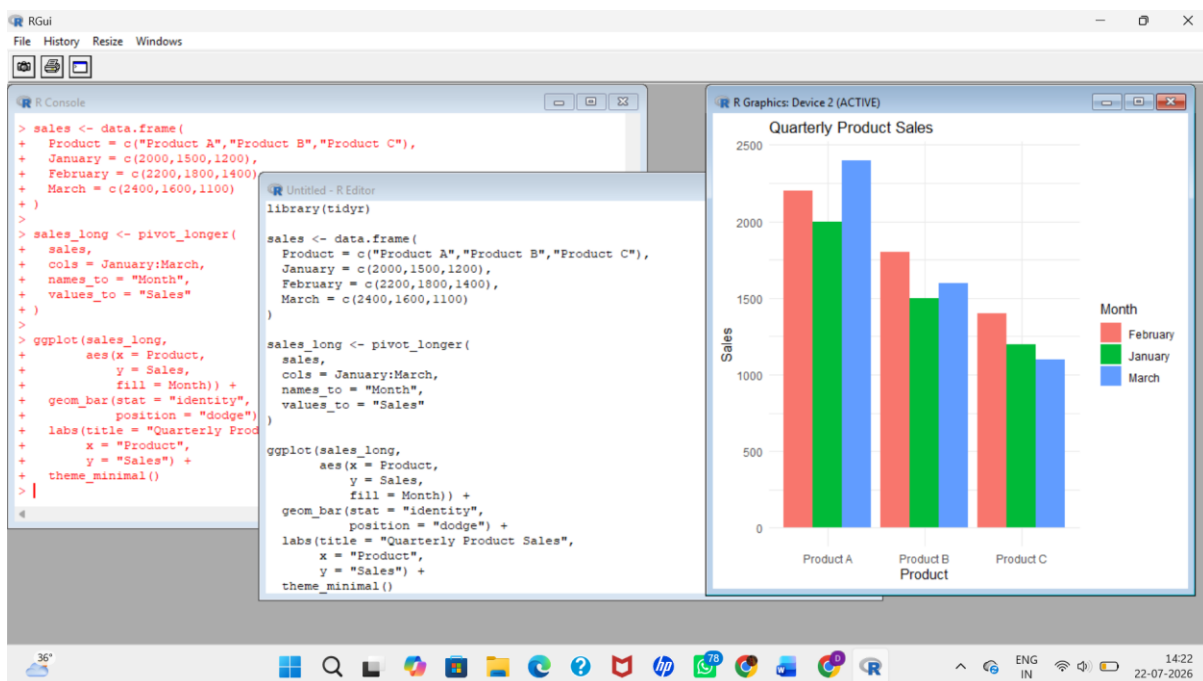
```
sales_long <- pivot_longer(
```

```
  sales,
```

```

cols = January:March,
names_to = "Month",
values_to = "Sales"
)
ggplot(sales_long,
  aes(x = Product,
      y = Sales,
      fill = Month)) +
geom_bar(stat = "identity",
  position = "dodge") +
labs(title = "Quarterly Product Sales",
  x = "Product",
  y = "Sales") +
theme_minimal()

```



Q2. Stacked Area Chart

```
library(ggplot2)
```

```
library(tidyr)
```

```

sales_long <- pivot_longer(
  sales,
  cols = January:March,
  names_to = "Month",
  values_to = "Sales"
)

ggplot(sales_long,
  aes(x = Month,
      y = Sales,
      fill = Product,
      group = Product)) +
  geom_area(alpha = 0.7) +
  labs(title = "Overall Sales Trend",
      x = "Month",
      y = "Sales") +
  theme_minimal()

```



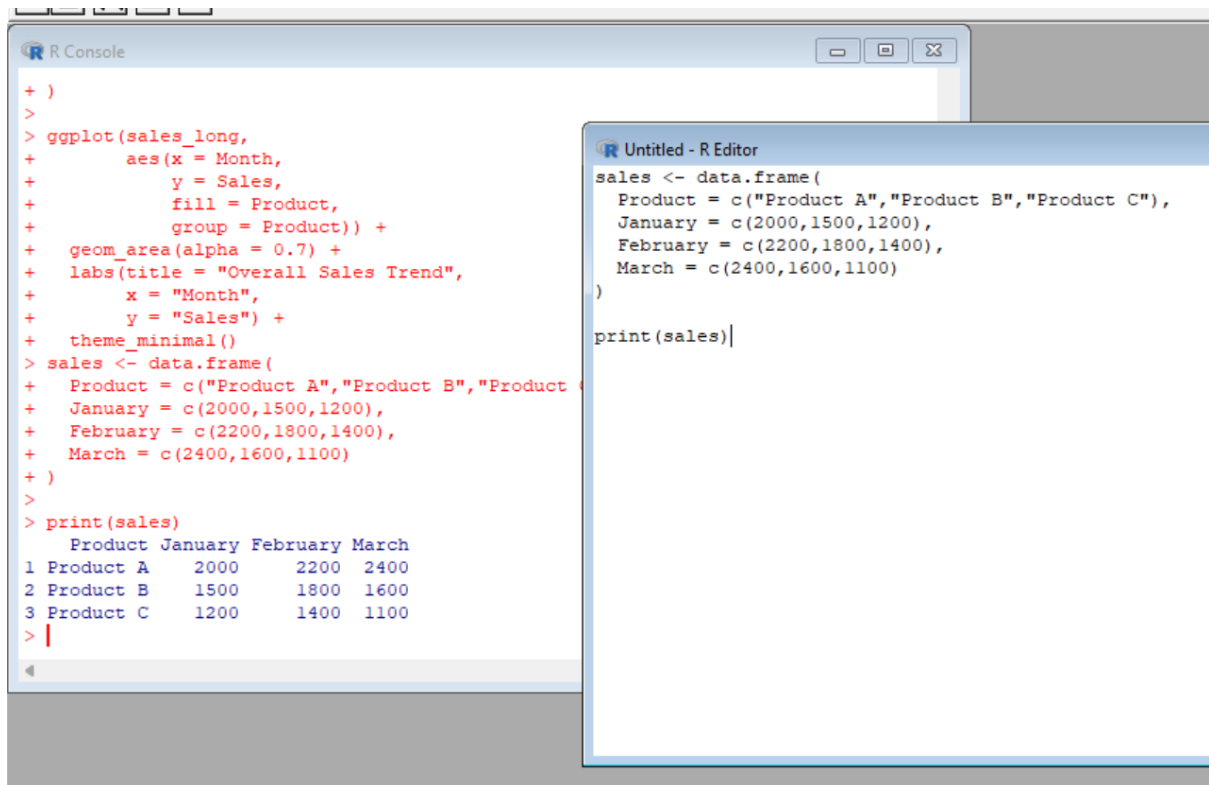
Q3. Table

```

sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

print(sales)

```



Q4. Dashboard

```

library(ggplot2)
library(gridExtra)
library(tidyr)

```

```

sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),

```

```
February = c(2200,1800,1400),  
March = c(2400,1600,1100)  
)
```

```
sales_long <- pivot_longer(  
  sales,  
  cols = January:March,  
  names_to = "Month",  
  values_to = "Sales"  
)
```

```
# Grouped Bar Chart
```

```
p1 <- ggplot(sales_long,  
  aes(Product, Sales, fill = Month)) +  
  geom_bar(stat = "identity",  
    position = "dodge") +  
  labs(title = "Grouped Bar Chart") +  
  theme_minimal()
```

```
# Stacked Area Chart
```

```
p2 <- ggplot(sales_long,  
  aes(Month, Sales,  
    fill = Product,  
    group = Product)) +  
  geom_area(alpha = 0.7) +  
  labs(title = "Stacked Area Chart") +  
  theme_minimal()
```

```
# Table
```

```
table_plot <- tableGrob(sales)
```

```
# Dashboard
```

```
grid.arrange(
```

```
  p1,
```

```
  p2,
```

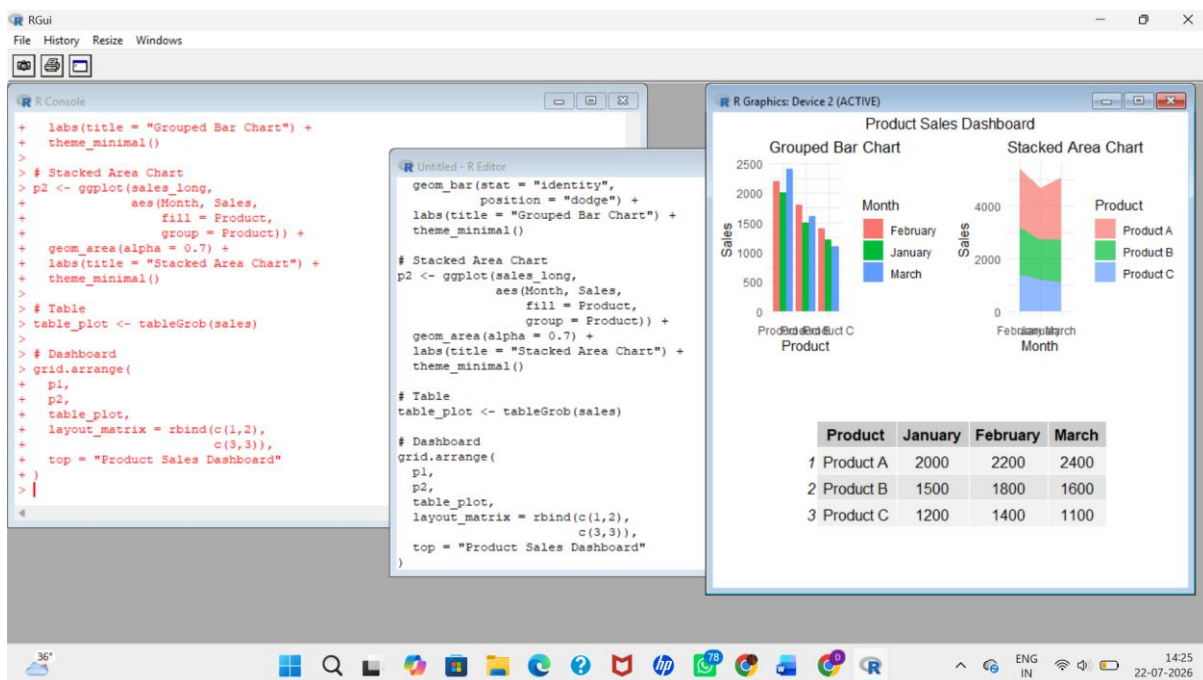
```
  table_plot,
```

```
  layout_matrix = rbind(c(1,2),
```

```
                        c(3,3)),
```

```
  top = "Product Sales Dashboard"
```

```
)
```



Set 16 – Customer Demographics Analysis

Q1. Bar Chart

```
library(ggplot2)
```

```
customers <- data.frame(
```

```
  Customer = c(1,2,3),
```

```
  Age = c(28,35,42),
```

```

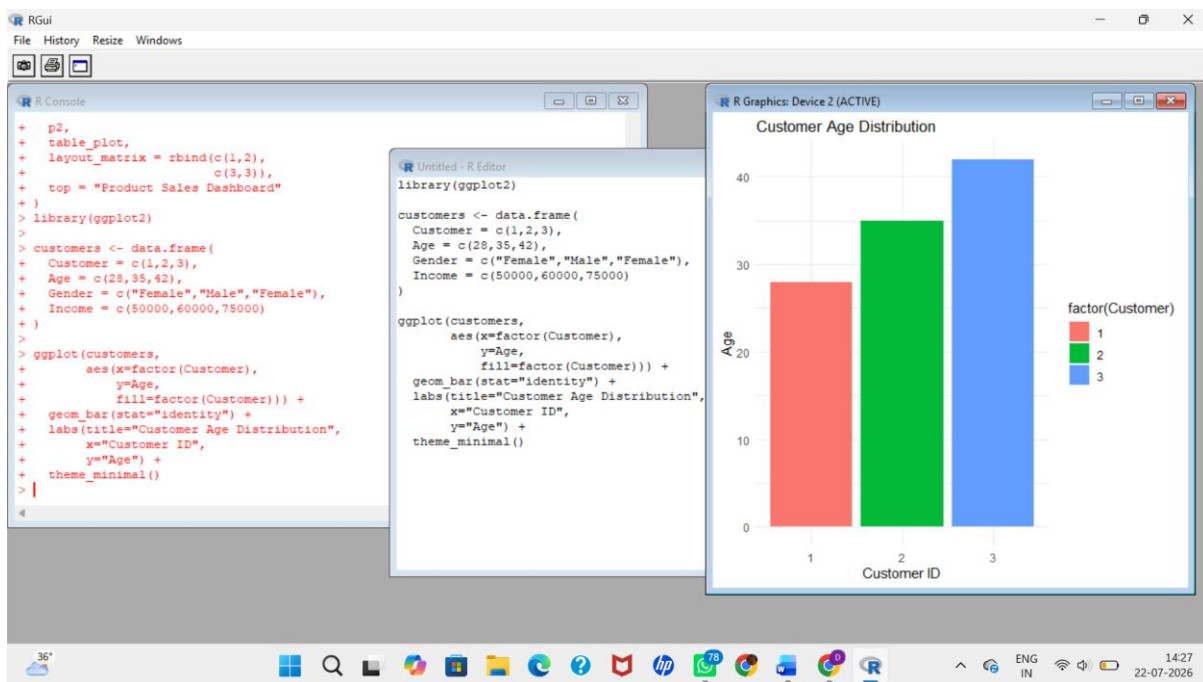
Gender = c("Female","Male","Female"),
Income = c(50000,60000,75000)
)

```

```

ggplot(customers,
  aes(x=factor(Customer),
      y=Age,
      fill=factor(Customer))) +
geom_bar(stat="identity") +
labs(title="Customer Age Distribution",
      x="Customer ID",
      y="Age") +
theme_minimal()

```



Q2. Pie Chart

```

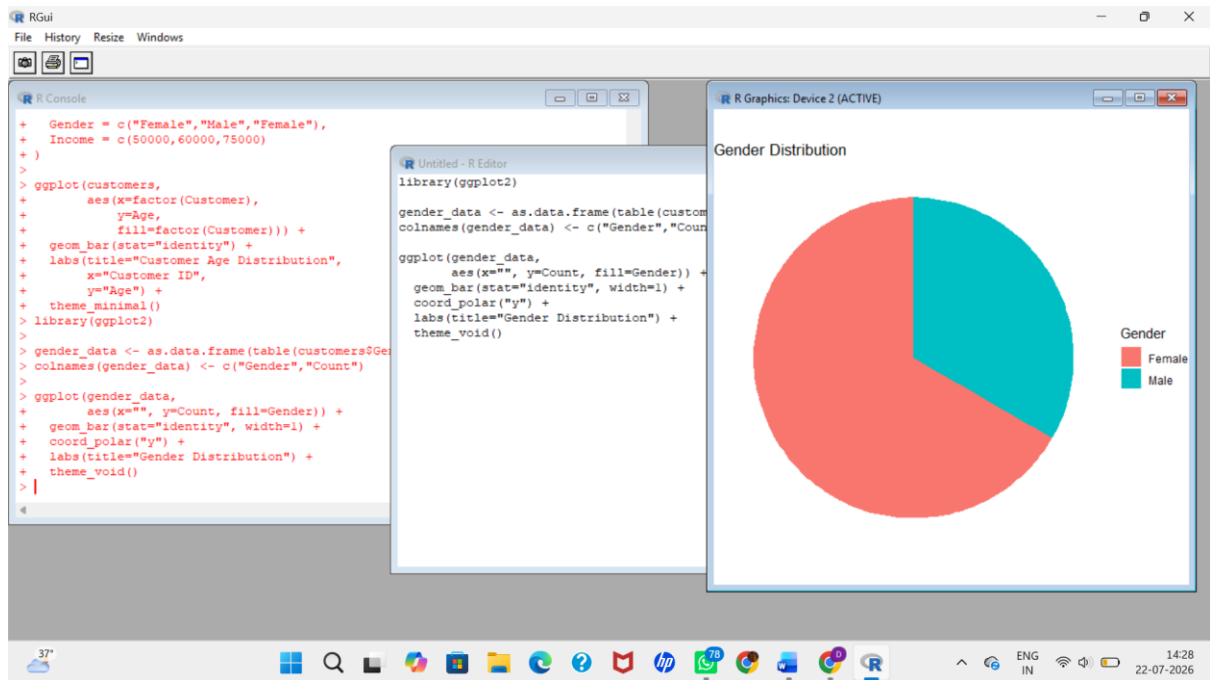
library(ggplot2)

gender_data <- as.data.frame(table(customers$Gender))

colnames(gender_data) <- c("Gender","Count")

```

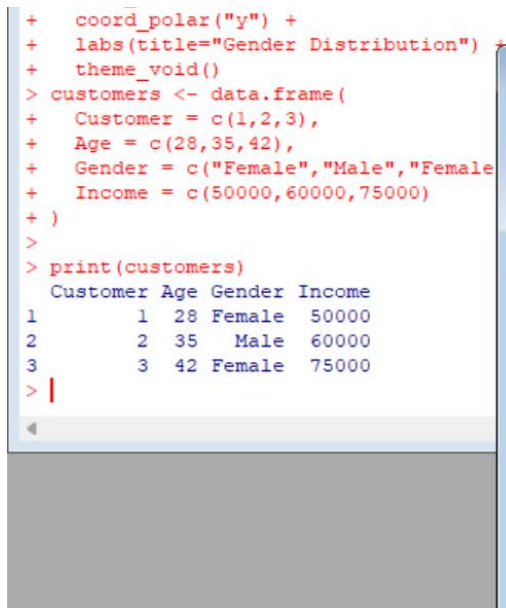
```
ggplot(gender_data,
      aes(x="", y=Count, fill=Gender)) +
geom_bar(stat="identity", width=1) +
coord_polar("y") +
labs(title="Gender Distribution") +
theme_void()
```



Q3. Table

```
customers <- data.frame(
  Customer = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
print(customers)
```

```
+ coord_polar("y") +
+ labs(title="Gender Distribution") +
+ theme_void()
> customers <- data.frame(
+   Customer = c(1,2,3),
+   Age = c(28,35,42),
+   Gender = c("Female","Male","Female"),
+   Income = c(50000,60000,75000)
+ )
>
> print(customers)
  Customer Age Gender Income
1         1  28 Female 50000
2         2  35  Male 60000
3         3  42 Female 75000
> |
```



```
Untitled - R Editor
customers <- data.frame(
  Customer = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
print(customers)
```

Q4: Tableau Dashboard

```
library(ggplot2)
```

```
library(gridExtra)
```

```
customers <- data.frame(
```

```
  Customer = c(1,2,3),
```

```
  Age = c(28,35,42),
```

```
  Gender = c("Female","Male","Female"),
```

```
  Income = c(50000,60000,75000)
```

```
)
```

```
p1 <- ggplot(customers,
```

```
  aes(x=factor(Customer),
```

```
    y=Age,
```

```
    fill=factor(Customer))) +
```

```
geom_bar(stat="identity") +
```

```
labs(title="Customer Age Distribution",
```

```
  x="Customer ID",
```

```
  y="Age") +
```

```
theme_minimal()
```

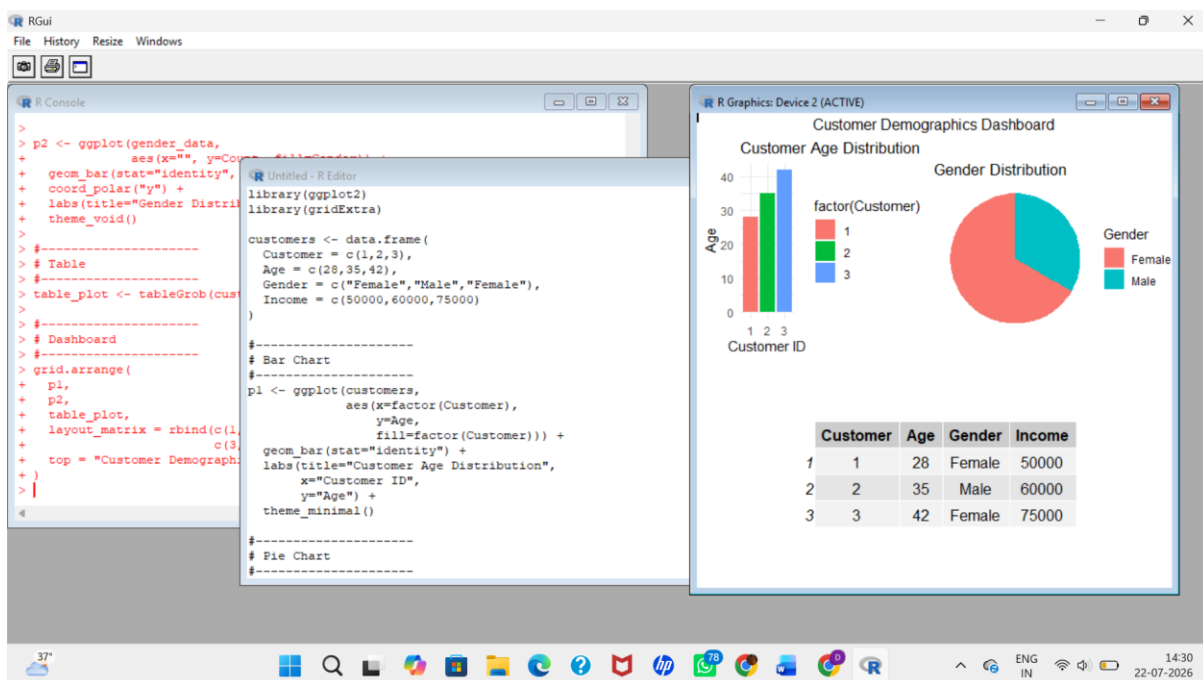
```
gender_data <- as.data.frame(table(customers$Gender))
```

```
colnames(gender_data) <- c("Gender", "Count")
```

```
p2 <- ggplot(gender_data,
             aes(x="", y=Count, fill=Gender)) +
  geom_bar(stat="identity", width=1) +
  coord_polar("y") +
  labs(title="Gender Distribution") +
  theme_void()
```

```
table_plot <- tableGrob(customers)
```

```
grid.arrange(
  p1,
  p2,
  table_plot,
  layout_matrix = rbind(c(1,2),
                        c(3,3)),
  top = "Customer Demographics Dashboard"
)
```



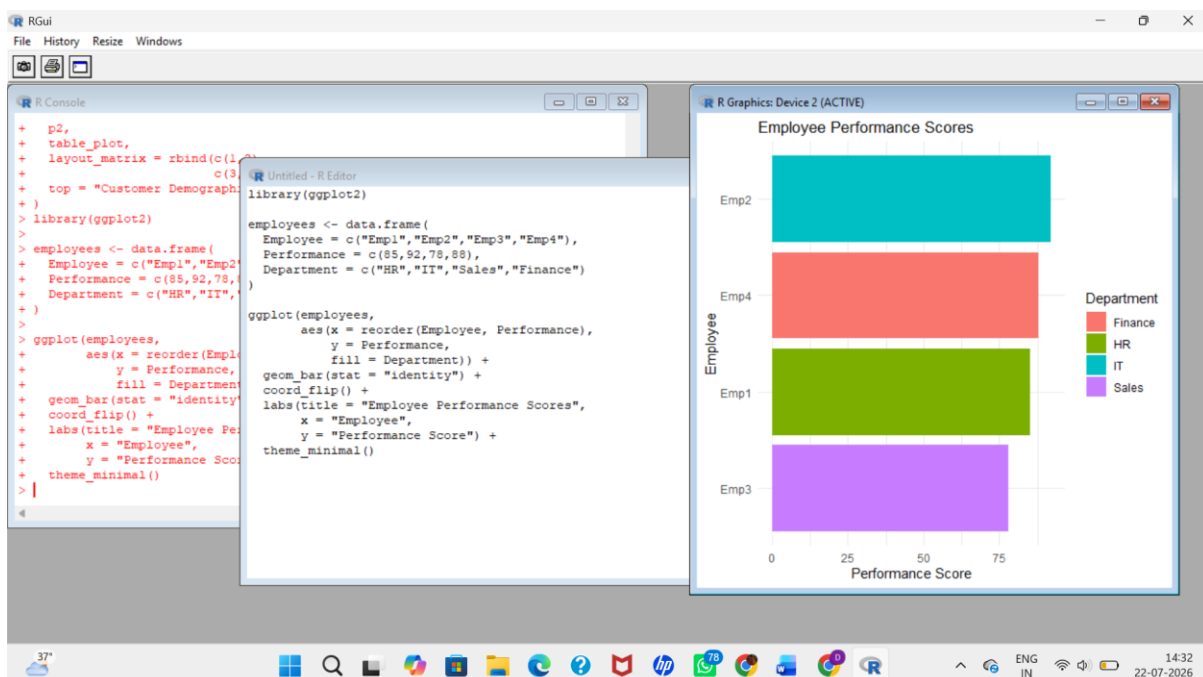
Set 17 – Employee Performance Analysis

Q1. Horizontal Bar Chart

```
library(ggplot2)
```

```
employees <- data.frame(  
  Employee = c("Emp1","Emp2","Emp3","Emp4"),  
  Performance = c(85,92,78,88),  
  Department = c("HR","IT","Sales","Finance")  
)
```

```
ggplot(employees,  
  aes(x = reorder(Employee, Performance),  
    y = Performance,  
    fill = Department)) +  
geom_bar(stat = "identity") +  
coord_flip() +  
labs(title = "Employee Performance Scores",  
  x = "Employee",  
  y = "Performance Score") +  
theme_minimal()
```

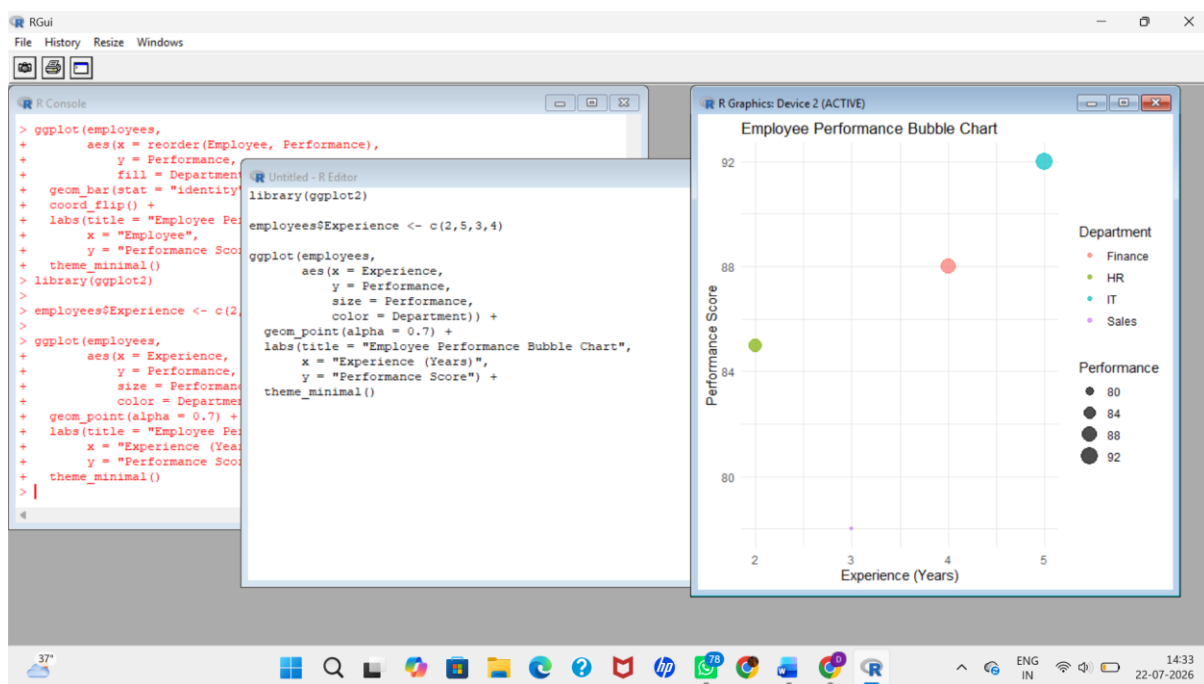


Q2. Bubble Chart

```
library(ggplot2)

employees$Experience <- c(2,5,3,4)

ggplot(employees,
       aes(x = Experience,
           y = Performance,
           size = Performance,
           color = Department)) +
  geom_point(alpha = 0.7) +
  labs(title = "Employee Performance Bubble Chart",
       x = "Experience (Years)",
       y = "Performance Score") +
  theme_minimal()
```



Q3. Table

```
employees <- data.frame(
  Employee = c("Emp1", "Emp2", "Emp3", "Emp4"),
  Performance = c(85, 92, 78, 88),
  Department = c("HR", "IT", "Sales", "Finance"),
```

```
Experience = c(2,5,3,4)
```

```
)
```

```
print(employees)
```

```
+ labs(title = "Employee Performance Bubble Chart",
+       x = "Experience (Years)",
+       y = "Performance Score") +
+ theme_minimal()
> employees <- data.frame(
+   Employee = c("Emp1", "Emp2", "Emp3", "Emp4"),
+   Performance = c(85, 92, 78, 88),
+   Department = c("HR", "IT", "Sales", "Finance"),
+   Experience = c(2, 5, 3, 4)
+ )
>
> print(employees)
  Employee Performance Department Experience
1    Emp1          85         HR           2
2    Emp2          92         IT           5
3    Emp3          78        Sales           3
4    Emp4          88        Finance          4
> |
```

Q4. Tableau Dashboard

```
library(ggplot2)
```

```
library(gridExtra)
```

```
employees <- data.frame(
```

```
  Employee = c("Emp1", "Emp2", "Emp3", "Emp4"),
```

```
  Performance = c(85, 92, 78, 88),
```

```
  Department = c("HR", "IT", "Sales", "Finance"),
```

```
  Experience = c(2, 5, 3, 4)
```

```
)
```

```
p1 <- ggplot(employees,
```

```
  aes(x = reorder(Employee, Performance),
```

```
      y = Performance,
```

```
      fill = Department)) +
```

```
geom_bar(stat = "identity") +
```

```
coord_flip() +
```

```
labs(title = "Employee Performance",
```

```
      x = "Employee",
```

```
      y = "Score") +
```

```
theme_minimal()
```

```

p2 <- ggplot(employees,
  aes(x = Experience,
      y = Performance,
      size = Performance,
      color = Department)) +
  geom_point(alpha = 0.8) +
  labs(title = "Bubble Chart") +
  theme_minimal()
table_plot <- tableGrob(employees)
grid.arrange(
  p1,
  p2,
  table_plot,
  layout_matrix = rbind(c(1,2),
                        c(3,3)),
  top = "Employee Performance Dashboard"
)

```

